COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

4 Complexity Theory (ad260)

For the purpose of this question, a graph $G = (V, E)$ is a set V of vertices along with a set E of edges where each edge is a set of two *distinct* vertices. That is, we consider undirected graphs without self-loops or multiple edges.

Given two graphs $G = (V, E)$ and $H = (U, F)$, a *homomorphism* from G to H is a function $h : V \to U$ such that whenever $\{v_1, v_2\}$ is in E, $\{h(v_1), h(v_2)\}$ is in F. We write HOM for the decision problem consisting of all pairs of graphs (G, H) such that there is a homomorphism from G to H.

Recall that a graph $G = (V, E)$ is k-colourable (for a positive integer k) if there is a function $\chi : V \to \{1, \ldots, k\}$ such that whenever $\{u, v\}$ is in E, $\chi(u) \neq \chi(v)$.

Non-determinism. NP.

(*a*) Explain why the decision problem HOM is in NP. [4 marks]

Answer: We can show that the problem is in NP because if we are given the input (G, H) along with the certificate $h : V \to U$, it is easy to verify that h satisfies the condition of being a homomorphism. Since the length of h is at most $|V|$ pairs of vertices, it is bounded by a polynomial in the length of the input and hence the problem is in NP.

Graph-theoretic Problems.

(*b*) Let K_3 denote the graph with three vertices a, b, c and the three edges $\{a, b\}, \{b, c\}$ and $\{a, c\}$. Show that for any graph G, there is a homomorphism from G to K_3 if, and only if, G is 3-colourable. [6 marks]

Answer: Suppose there is a function $h : V \to \{a, b, c\}$ which is a homomophism from $G = (V, E)$ to K_3. Then, for each edge $\{u, v\} in E$, there is an edge $\{h(u), h(v)\}$ in K_3. In particular, this implies that $h(u) \neq h(v)$. Thus, if we define a function $\chi : V \to \{1, 2, 3\}$ from h by identifying a with 1, b with 2 and c with 3, we see that G is 3-colourable.

In the other direction, suppose G is 3-colourable and this is witnessed by a function χ. Again, define a function $h : V \to \{a, b, c\}$ by composing χ with the bijection that takes 1 to a, 2 to b and 3 to c. To see that this is a homomorphism from G to K_3, note that whenever $\{u, v\} in E$, we have $\chi(u) \neq \chi(v)$ and so $h(u) \neq h(v)$. But for each pair $x, y \in \{a, b, c\}$ with $x \neq y$, there is an edge $\{x, y\}$ in K_3.

Reductions. NP-completeness.

(*c*) What can you conclude from the above about the complexity of the problem HOM? [5 marks]

Answer: The problem HOM is NP-complete. The fact that it is in NP is established in part (a). To show that it is NP-hard, we give a reduction from 3-colourability, which was shown to be NP-complete in the lecture notes.

The reduction takes a graph G to the pair (G, K_3). By part (b), we know that (G, K_3) is in HOM if, and only if, G is 3-colourable. Thus, this is a valid reduction. It is also trivially seen to be in polynomial time, since it only involves appending the graph K_3 to the input.

Polynomial-time problems and algorithms.

(*d*) Let K_2 denote the graph consisting of two vertices a and b and the single edge

$\{a, b\}$. What is the complexity of the decision problem consisting of all graphs G for which there is a homomorphism from G to K_2? [5 marks]

Answer: By an argument entirely analogous to part (b), we can show that there is a homomorphism from a graph G to K_2 if, and only if, G is 2-colourable. It is known that 2-colourability of graphs is decidable in polynomial time (indeed, even in logarithmic space, but this is not needed for full marks). So, the problem is in P.
